

# Air Conditioners Data Analysis

## Stage 6: Data Analysis

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

### Stage Objective

Stage 6 performs descriptive analysis on the validated product data and the summary tables produced earlier.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path = "/content/Air_Conditioners_Cleaned.csv"
df = pd.read_csv(csv_path)

print("CSV loaded successfully.")
print("Dataset shape:", df.shape)
```

CSV loaded successfully.  
Dataset shape: (720, 14)

### What Analysis Means Here

The project uses descriptive analysis: we measure and compare what is present in the supplied dataset. We do not claim that an observed association proves causation.

```
nums=df.select_dtypes(include=np.number).columns.tolist(); print(nums)

['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
```

### Reload the CSV

The dataset is loaded directly so the analysis notebook is independently reproducible.

```
print(df[nums].describe().T if nums else 'No numeric columns found.')
```

|                  | count | mean         | std          | min           | \ |
|------------------|-------|--------------|--------------|---------------|---|
| ratings          | 720.0 | 3.888194     | 0.612151     | 1.000000      |   |
| no_of_ratings    | 720.0 | 151.262500   | 562.513005   | 1.000000      |   |
| discount_price   | 720.0 | 40183.979167 | 11938.526137 | 199.000000    |   |
| actual_price     | 720.0 | 55479.741139 | 18984.776512 | 499.000000    |   |
| discount_amount  | 720.0 | 15295.761972 | 13102.876643 | -38541.000000 |   |
| discount_percent | 720.0 | -2.351720    | 262.575704   | -5938.520801  |   |
| tonnage          | 626.0 | 1.468164     | 0.501664     | 0.750000      |   |
| energy_star      | 597.0 | 3.577889     | 0.951534     | 1.000000      |   |

  

|                  | 25%          | 50%          | 75%          | max           |
|------------------|--------------|--------------|--------------|---------------|
| ratings          | 3.900000     | 4.000000     | 4.000000     | 5.000000      |
| no_of_ratings    | 8.000000     | 18.000000    | 40.000000    | 9577.000000   |
| discount_price   | 37282.500000 | 39190.000000 | 41999.250000 | 128800.000000 |
| actual_price     | 51472.500000 | 56990.000000 | 61990.000000 | 149000.000000 |
| discount_amount  | 12726.750000 | 17800.000000 | 20403.250000 | 65000.000000  |
| discount_percent | 25.013559    | 31.23355     | 34.526545    | 60.12024      |
| tonnage          | 1.000000     | 1.50000      | 1.500000     | 8.50000       |
| energy_star      | 3.000000     | 3.00000      | 5.000000     | 5.00000       |

### Overall Statistics

Descriptive statistics are generated for numerical fields to understand typical values and the observed spread.

```
print(df['discount_price'].describe() if 'discount_price' in df.columns else 'Price field not detected.')
```

|       |               |
|-------|---------------|
| count | 720.000000    |
| mean  | 40183.979167  |
| std   | 11938.526137  |
| min   | 199.000000    |
| 25%   | 37282.500000  |
| 50%   | 39190.000000  |
| 75%   | 41999.250000  |
| max   | 128800.000000 |

Name: discount\_price, dtype: float64

### Price Analysis

The available price field is **discount\_price**. Its distribution, range and summary statistics are examined.

```
print(df['ratings'].describe() if 'ratings' in df.columns else 'Rating field not detected.')
```

|       |            |
|-------|------------|
| count | 720.000000 |
| mean  | 3.888194   |
| std   | 0.612151   |
| min   | 1.000000   |
| 25%   | 3.900000   |
| 50%   | 4.000000   |
| 75%   | 4.000000   |
| max   | 5.000000   |

Name: ratings, dtype: float64

## Rating Analysis

The available rating field is **ratings**. Its range and distribution are examined.

```
cats=df.select_dtypes(include=['object','category','bool']).columns.tolist()
for c in cats[:5]: print(c,df[c].value_counts().head(10).to_dict())

name {'Samsung 1.5 Ton 3 Star Wind-Free Technology Inverter Split AC (Copper, Convertible 5-in-1 Cooling Mode, Anti-bacterial Fil...': 3, 'White Westinghouse 1.5
main_category {'appliances': 720}
sub_category {'Air Conditioners': 720}
image {'https://m.media-amazon.com/images/I/51JFb7FctDL._AC_UL320_.jpg': 6, 'https://m.media-amazon.com/images/I/417b-AztnjL._AC_UL320_.jpg': 5, 'https://m.media
link {'https://www.amazon.in/Portable-Conditioner-Personal-Evaporative-Display/dp/B09YRMFZGJ/ref=sr_1_70?qid=1679134274&s=kitchen&sr=1-70': 1, 'https://www.am
```

## Category Comparisons

Group-level results are compared to understand how product attributes differ between available categories.

```
for c in nums[:5]: print('Top',c); print(df[[c]].sort_values(c,ascending=False).head(5))
```

```
Top ratings
ratings
460      5.0
674      5.0
451      5.0
423      5.0
58       5.0
Top no_of_ratings
no_of_ratings
237      9577.0
12       5073.0
14       3670.0
33       3604.0
245      3562.0
Top discount_price
discount_price
460      128800.0
421      102800.0
388       99750.0
666       87800.0
285       82260.0
Top actual_price
actual_price
460      149000.0
582      130000.0
388      128900.0
421      120490.0
583      119000.0
Top discount_amount
discount_amount
582       65000.0
583       59500.0
550       51000.0
28        43510.0
5         39000.0
```

## High and Low Observations

Sorted results are used to inspect high and low values and connect the numerical summaries to concrete product records.

```
print(df[nums].corr() if len(nums)>=2 else 'Need at least two numeric fields for correlation.')
```

```
ratings      ratings  no_of_ratings  discount_price  actual_price \
no_of_ratings  0.058132    1.000000    -0.059465    -0.050781
discount_price 0.018044   -0.059465    1.000000    0.730784
actual_price   0.170766   -0.050781    0.730784    1.000000
discount_amount 0.230984   -0.019396    0.147696    0.783056
discount_percent 0.070216  -0.037628   -0.004204    0.333476
tonnage       -0.003417   -0.073532    0.463141    0.298213
energy_star    0.079045    0.106380    0.178746    0.202723

ratings      discount_amount  discount_percent  tonnage  energy_star
no_of_ratings  -0.019396    -0.037628  -0.073532    0.106380
discount_price 0.147696    -0.004204  0.463141    0.178746
actual_price   0.783056    0.333476  0.298213    0.202723
discount_amount 1.000000    0.487004  -0.085923    0.106136
discount_percent 0.487004    1.000000  -0.565622    0.026140
tonnage       -0.085923    -0.565622  1.000000   -0.050691
energy_star    0.106136    0.026140  -0.050691    1.000000
```

## Relationship Checks

Where two numerical fields support a meaningful comparison, their relationship is examined descriptively through correlation or paired values.

```
print('Descriptive comparisons completed.')
```

```
Descriptive comparisons completed.
```

## What We Learned

The output of this stage is a set of measurable patterns that can be communicated more clearly through visualisations.

```
print('Stage 6 findings are ready for visualization.')
```

Stage 6 findings are ready for visualization.